



Physics-Informed Machine Learning for Cross-Surface Generalization in Droplet Impact Prediction

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1 The question

Can a model trained on 12 laser-textured surfaces predict how far a droplet spreads on a surface it has never seen?

Models of droplet impact are usually scored on random train/test splits, where every surface appears in training. Surface design needs the opposite: predictions for a surface **before** it is made and tested.

1,498 impacts on 12 textured AI surfaces	125 impacts on smooth REF-H (held out)	1–160 mPa·s: water–glycerol, 0–91 wt. %
8–119 Weber number	9–4435 Reynolds number	50–800 μm channel pitch; depth 6 or 26 μm

Open dataset of Može et al. [1]: FDP A-coated aluminium 1050A, square grids of laser channels, 5 000 fps imaging. Target: maximum spreading factor $\beta_{\max} = D_{\max}/D_0$, the one outcome measured on both surface types.

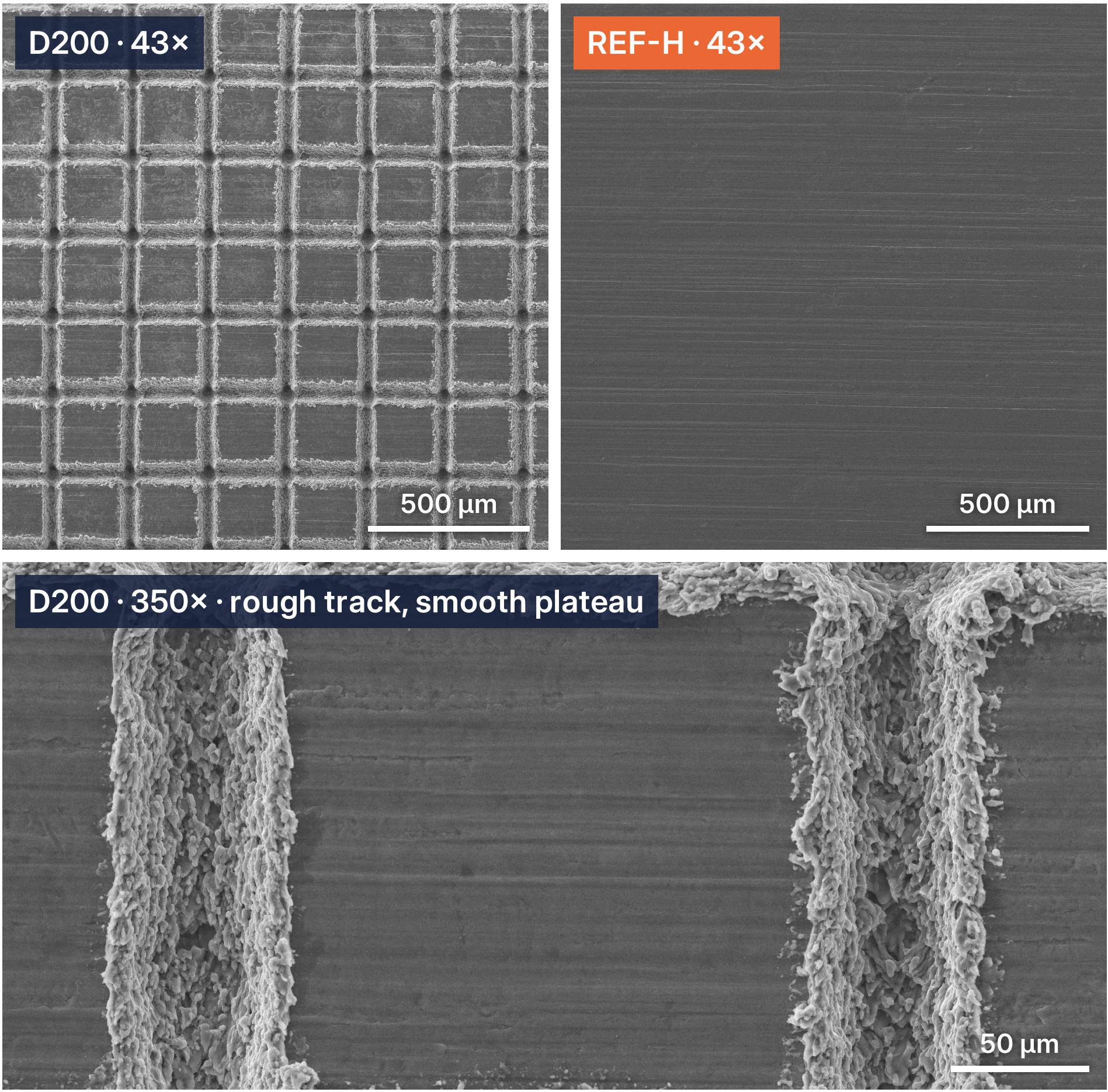
2 Physics inputs

$$Re = \frac{\rho V D_0}{\mu} \quad We = \frac{\rho V^2 D_0}{\sigma} \quad Oh = \frac{We}{Re}$$

Recomputed Re and We match the dataset to 5×10^{-7} . Because Oh is a function of Re and We , the dimensionless set carries two groups, not three.

Classic scaling laws fit this superhydrophobic data poorly: Laan et al. [3] gives $R^2 = 0.71$. They were derived on wettable surfaces, so we use them as a check, not as the model.

3 Reading the surface from SEM



The plateaus between laser tracks are the **same smooth surface as REF-H** [1]. Segmenting 43× images by local texture gives track widths $w \approx 45 \mu\text{m}$ (deep) and $30 \mu\text{m}$ (shallow), so every surface gets a smooth-plateau fraction:

$$\varphi = (\max(s - w, 0) / s)^2, \quad V_{\text{tex}} = (1 - \varphi) h$$

Textured surfaces span $\varphi = 0.01$ (D50) to 0.93 (S800). **REF-H is simply $\varphi = 1$:** a physical value, not a made-up spacing or depth.

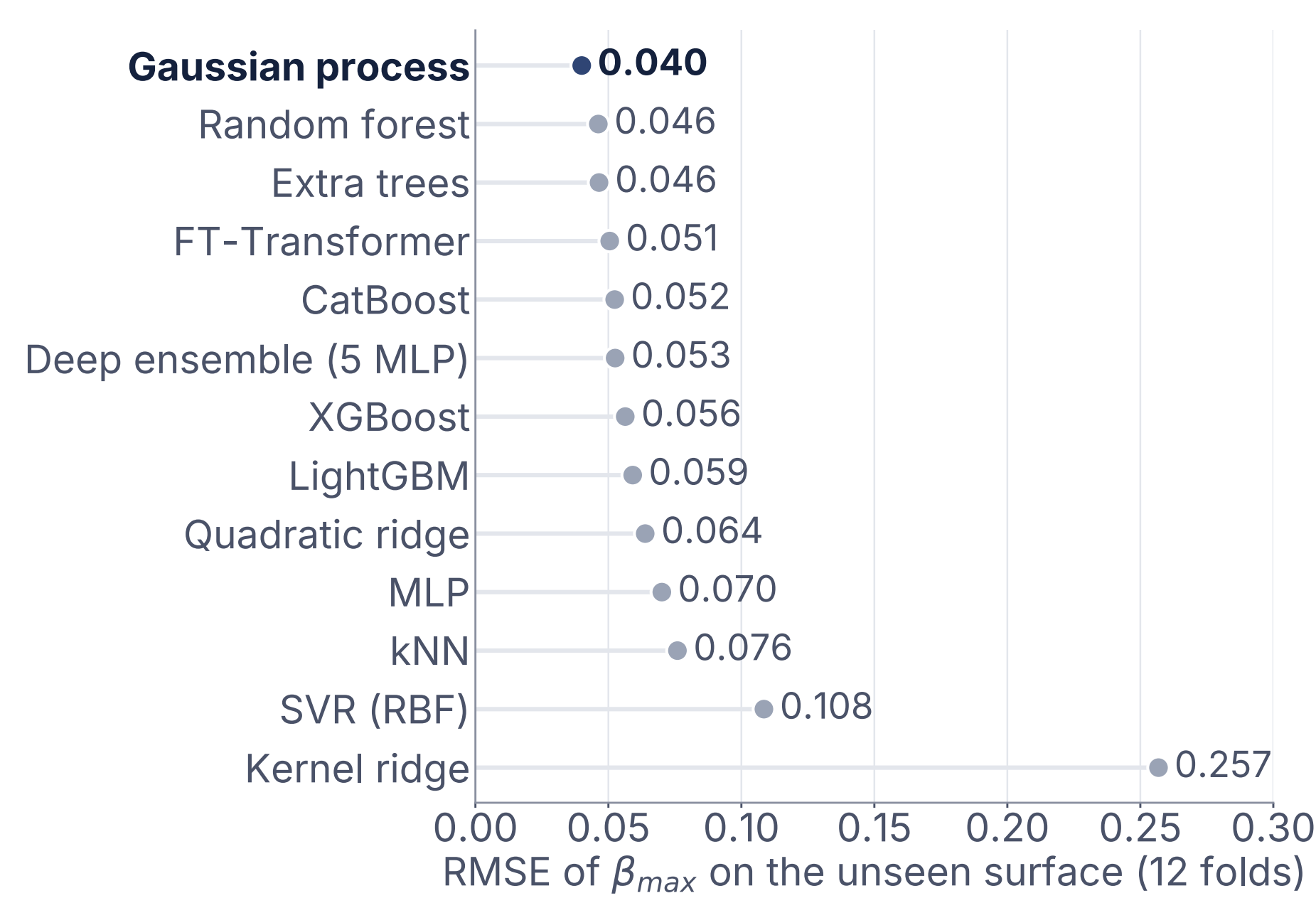
4 Leakage-safe evaluation

Each condition was repeated 5×. A random split puts near-identical twins in train and test and flatters every model.

- Grouped 5-fold CV** over 297 replicate groups: in-distribution accuracy.
- Leave-one-surface-out**, 12 folds: the model-selection criterion.
- REF-H, predicted once** by models trained on all textured data. Never used for tuning.

All models get the same inputs: $\log Re$, $\log We$, $\log D_0$, φ , V_{tex} . Target: $\log \beta_{\max}$.

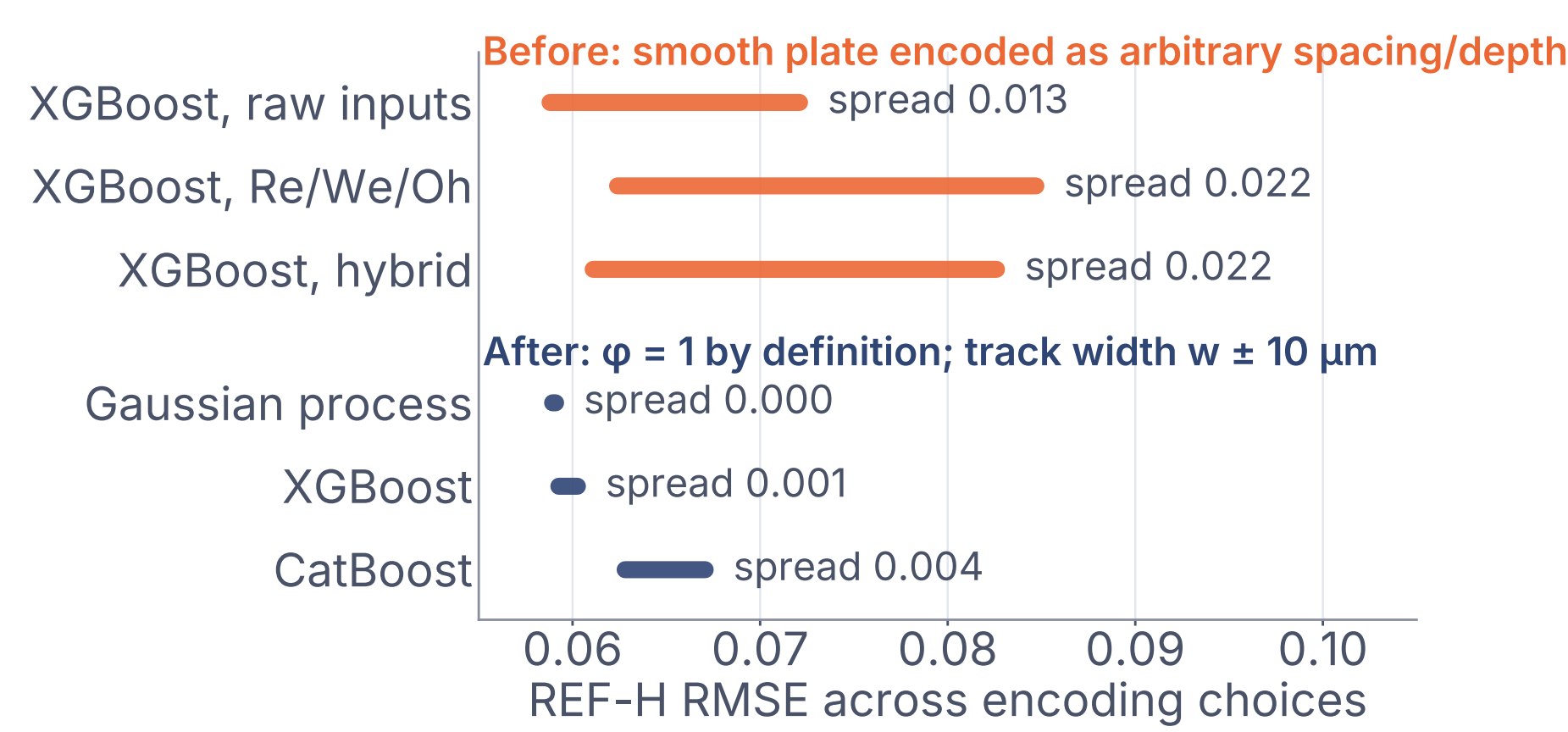
5 13 models, one test



Gaussian process regression (Matérn 3/2, one length-scale per input) generalizes best to unseen surfaces: 29% lower error than XGBoost on the same inputs. Kernel ridge and SVR fit well in-distribution (≈ 0.045) but collapse on unseen surfaces. Every model's worst surface is D50, the densest texture.

0.036 GPR RMSE, in-distribution ($R^2 = 0.993$)	0.040 GPR RMSE, unseen textured surface	0.056 XGBoost, unseen textured surface
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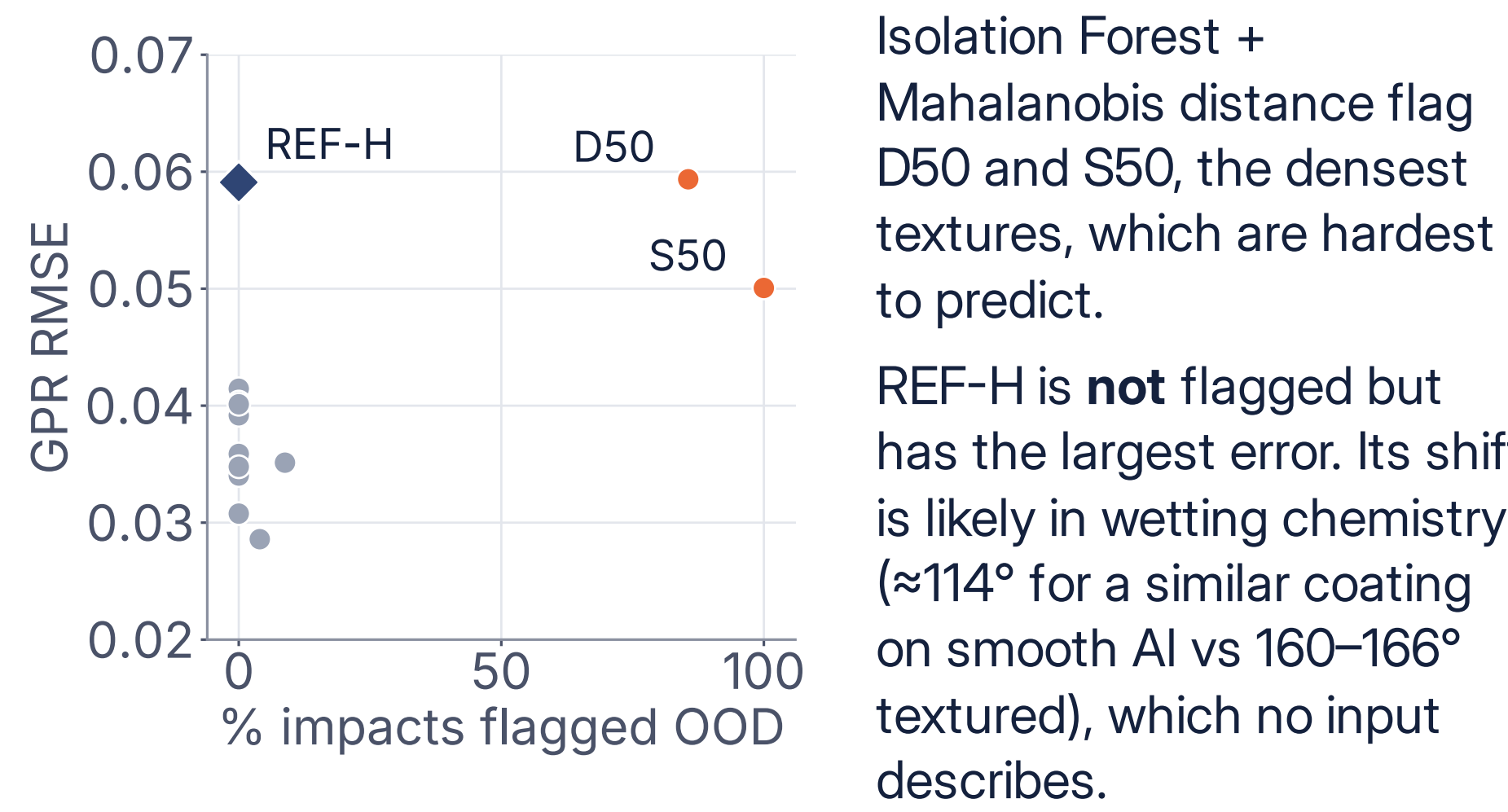
6 Why the descriptor matters



A smooth plate has no channel spacing or depth. Encoding REF-H with invented values (0/0, 50/6, 800/25 μm...) moved its error by up to **0.022**. With $\varphi = 1$, changing the measured track width by $\pm 10 \mu\text{m}$ moves it by at most **0.005**.

Geometry is not negligible: removing surface inputs nearly doubles in-distribution error (0.086 vs 0.051). Deep, dense textures spread further.

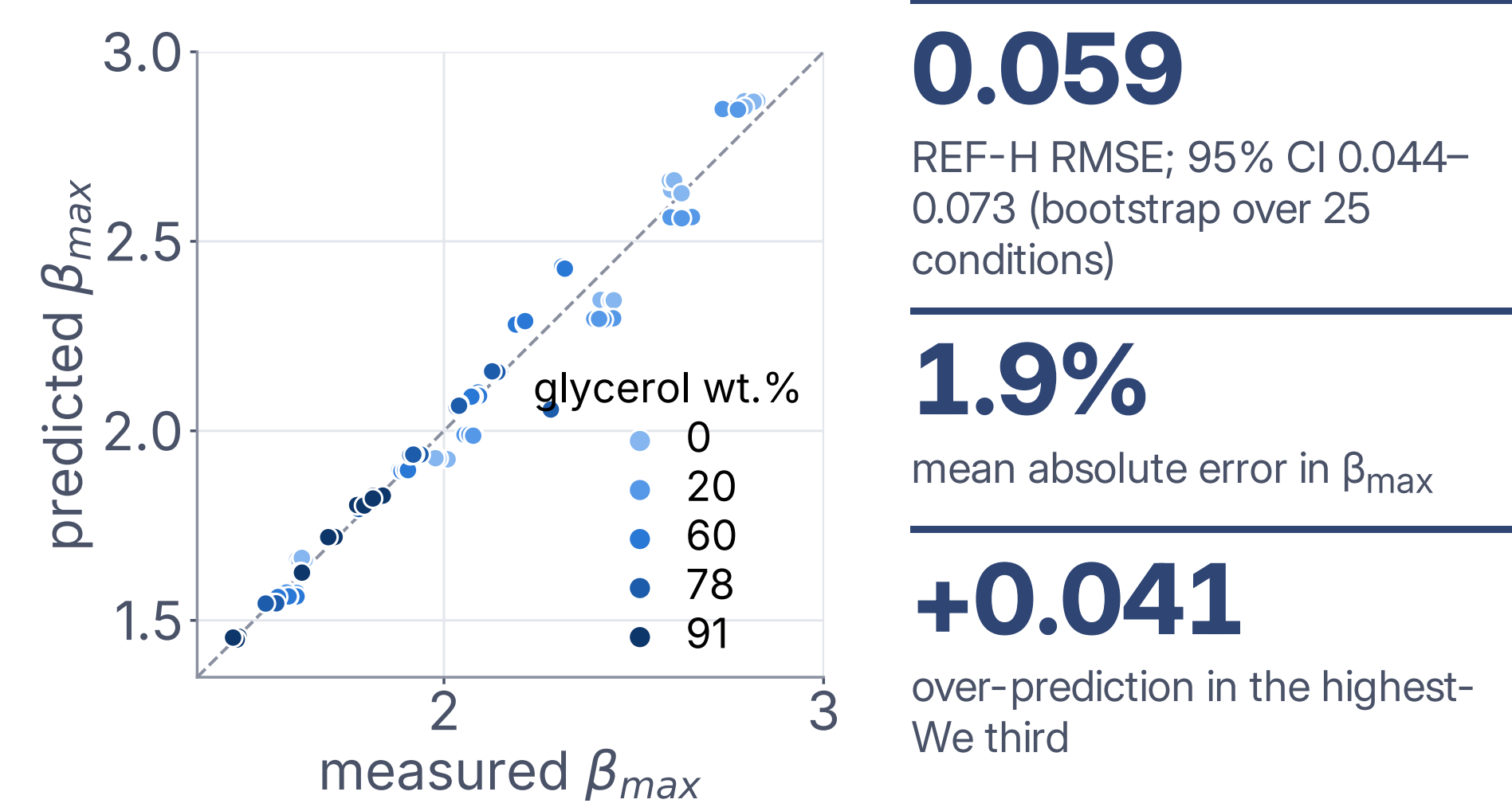
7 Knowing when not to trust it



Isolation Forest + Mahalanobis distance flag D50 and S50, the densest textures, which are hardest to predict.

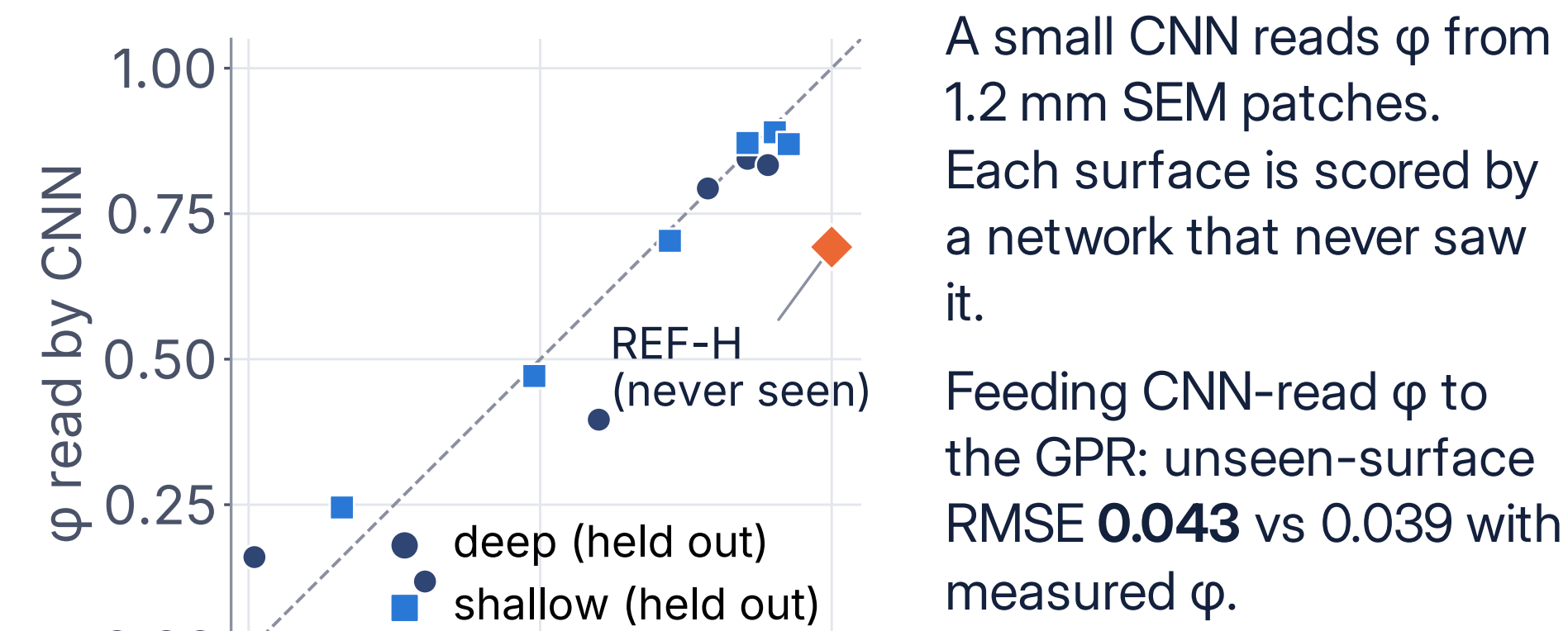
REF-H is **not** flagged but has the largest error. Its shift is likely in wetting chemistry ($\approx 114^\circ$ for a similar coating on smooth Al vs $160\text{--}166^\circ$ textured), which no input describes.

8 Blind test on the smooth surface



Spreading transfers well because it is inertia-driven. The residual error sits at high We , where textured surfaces spread further than the smooth plate. Possible cause: less friction over the air cushion **HYPOTHESIS**

9 From an SEM image to a prediction



REF-H is read as $\varphi = 0.69$ rather than 1.0, because no training surface is smoother than $\varphi = 0.93$. Image-derived descriptors work within the texture family they were trained on.

10 The virtual droplet lab

The pieces fit together as one tool: enter a fluid, an impact speed and an SEM image of a new surface, and get $\beta_{\max}$ with an interval and a trust flag.

Inputs	Physics	GPR
D_0 , V , ρ , σ , μ + SEM image	Re , We ; CNN reads φ , V_{tex}	$\beta_{\max}$ and predictive σ
OOD check	Regime flags	Tested
Isolation Forest + Mahalanobis	$Re < 70$: outside scaling-law range	unseen surfaces + blind REF-H

★ Take-aways

- Judge droplet models on **unseen surfaces**. Random splits hide the gap.
- Describe surfaces with quantities every surface has. The smooth-plateau fraction φ makes the smooth surface a limit of the textured family.
- A **Gaussian process** on physics inputs beats 12 alternatives, including boosted trees and a transformer, and predicts the smooth surface to **1.9%**.

Next: a dynamic contact-angle descriptor (β_0 correction of Lee et al. [4]); calibrated conformal intervals; rebound and contact-time targets.

1. M. Može et al., *Data in Brief* 61 (2025) 111697; data doi:10.17632/wsh8rxwd38.1
2. ML study on this dataset, *Biomimetics* 10 (2025) 357.
3. N. Laan et al., *Phys. Rev. Applied* 2 (2014) 044018.
4. J.B. Lee et al., *J. Fluid Mech.* 786 (2016) R4.